

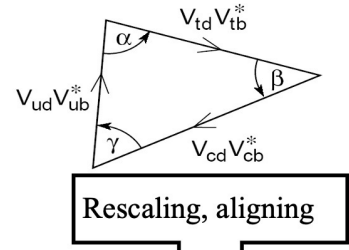
CP Violation in B mesons

$$V_{CKM} = \begin{pmatrix} 1 - \lambda^2/2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \lambda^2/2 & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} + \mathcal{O}(\lambda^4)$$

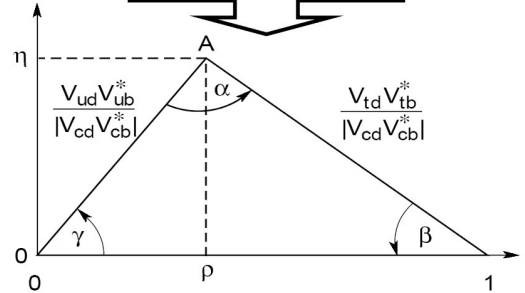
V_{td}

complex phase.

V_{ub}



Rescaling, aligning

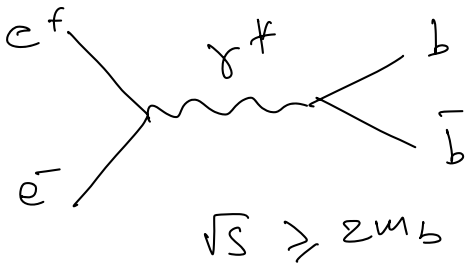


Experimentally prove $\alpha \neq 0$
 $\beta \neq 0$
 $\gamma \neq 0$

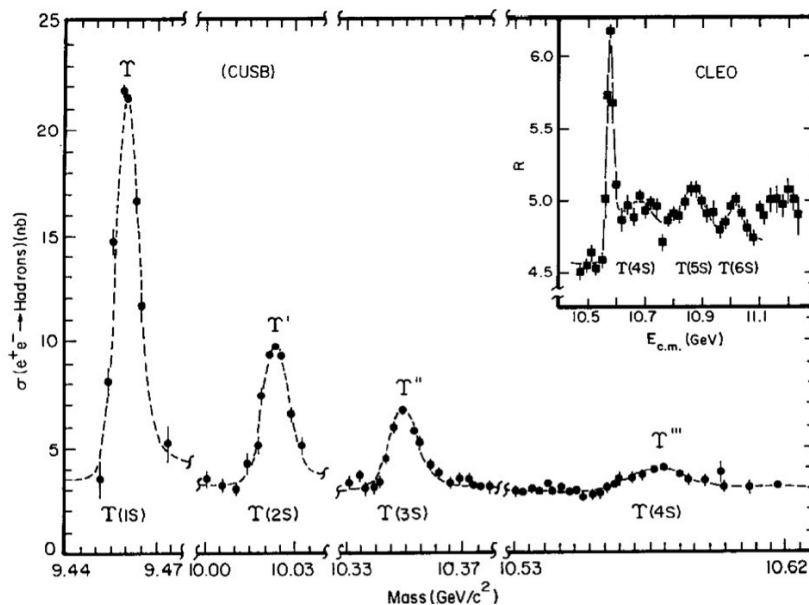
$\Rightarrow$ produce B mesons and study their decays

Direct CP: $B^- \rightarrow f \neq \bar{B}^- \rightarrow \bar{f}$?

Interference: $B^0 \rightarrow \bar{B}^0 \rightarrow f_{CP} + B^0 \rightarrow f_{CP}$ account for interf
 f_{CP} : CP eigenstate



$e^+e^- \rightarrow \Upsilon(nS) \rightarrow \text{hadrons}$



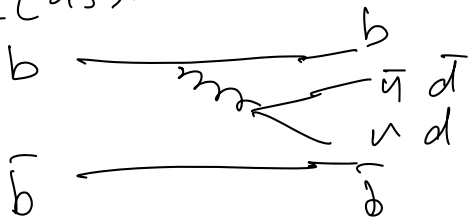
$$\sqrt{s} \approx m_{\Upsilon(4S)} \approx 10.58 \text{ GeV}$$

$$m_{\Upsilon(1S)} < 2 \times m_B$$



$\Upsilon(1S)$ decay through 3-gluon exchange.

$\Upsilon(4S)$.



$$\Upsilon(4S) \rightarrow B^0 \bar{B}^0$$

$$B^+ B^-$$

$$Q = m_{\Upsilon(4S)} - 2 m_B$$

$$= 10.58 \text{ GeV} - 2 \times 5.28 \text{ GeV}$$

$$\sqrt{s} = 10.58 \text{ GeV}$$

$$\Upsilon(4S) \rightarrow B^0 \bar{B}^0 \quad \text{not allowed Kine.}$$

$$e^+ e^- \rightarrow$$

$$\mu^+ \mu^-$$

$$\tau^+ \tau^-$$

$$u \bar{u}$$

$$d \bar{d}$$

$$c \bar{c}$$

$$s \bar{s}$$

$$b \bar{b}$$

Kinematically allowed

$$\frac{\sigma(e^+ e^- \rightarrow b \bar{b})}{\sigma(e^+ e^-)_{\text{TOT}}} \approx 25\%$$

$$BF(\Upsilon(4S) \rightarrow B^0 \bar{B}^0) \approx 50\%$$

$$BF(\Upsilon(4S) \rightarrow B^+ B^-)$$

B-Factory

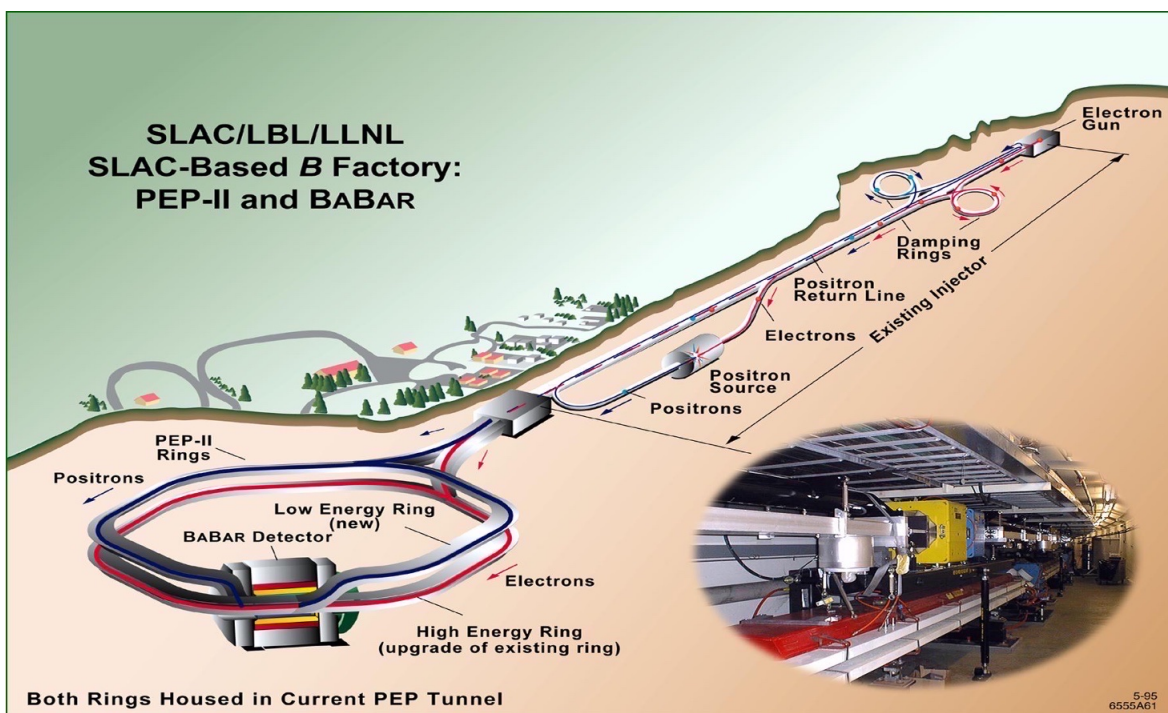
SLAC

PEP-II

BaBar

KEK
Japan.

BELLE



asymm. beams

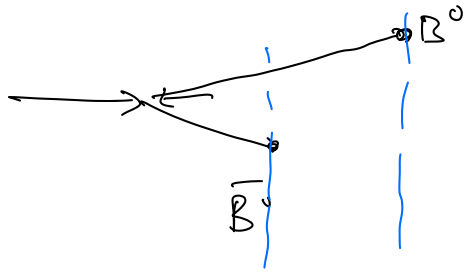
$$9.1 \text{ GeV } e^- \rightarrow \leftarrow e^+ 3 \text{ GeV}$$

$$Q < m_{\pi}$$

$$m_B \approx 5.279$$

$$P_B^* \approx 200 \text{ MeV} \ll m_B$$

$$\text{Boost } \beta\gamma = 0.56$$



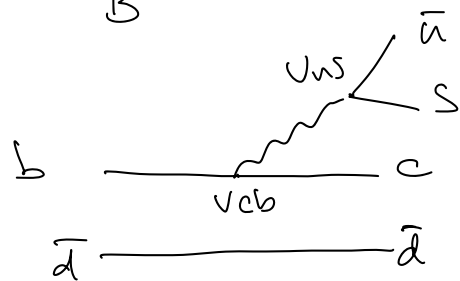
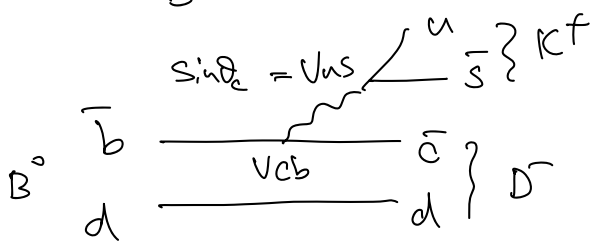
$$\langle \ell \rangle \approx \beta\gamma c \tau_B \approx 150 \mu\text{m}$$

Direct CP

$$B^0 \rightarrow D^- K^+$$

$$\stackrel{CP}{\Rightarrow}$$

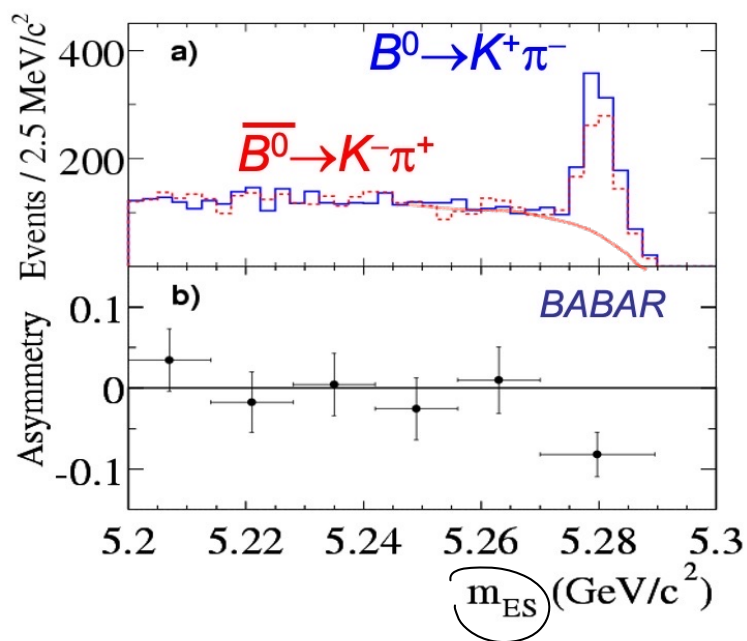
$$\bar{B}^0 \rightarrow D^+ K^-$$



$$N(B^0 \rightarrow D^- K^+) \propto |V_{us}|^2 |V_{cb}|^2$$

First collisions: May 1999

$$A_{K^-\pi^+} \equiv \frac{\Gamma(\bar{B} \rightarrow K^-\pi^+) - \Gamma(B \rightarrow K^+\pi^-)}{\Gamma(\bar{B} \rightarrow K^-\pi^+) + \Gamma(B \rightarrow K^+\pi^-)}$$



$$n_{K\pi} = 1606 \pm 51$$

$$A_{K\pi} = -0.133 \pm 0.030 \pm 0.009$$

$$n(B^0 \rightarrow K^+\pi^-) = 910$$

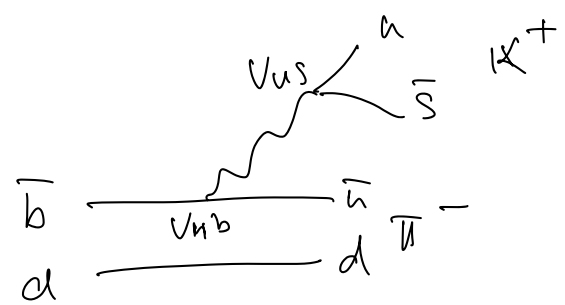
$$n(\bar{B}^0 \rightarrow K^-\pi^+) = 696$$

2004

$$m_{ES} = m_{K\pi}$$

$$\#(B \rightarrow K\pi) \propto |V_{us}|^2 |V_{ub}|^2$$

$$\#(B \rightarrow DK) \propto |V_{cb}|^2 |V_{us}|^2$$



CP violation in interference

CP eigenstates accessible to B^0 and $\bar{B}^0$

$$f_{CP} = J/\psi \quad K_S^0 \quad K_S^0 \rightarrow \pi^+ \pi^-$$

$$L \rightarrow e^+ e^-$$

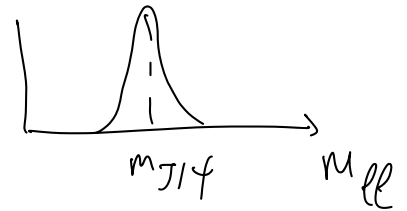
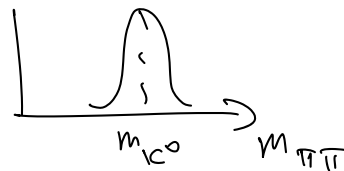
Final state $e^+ e^- \pi^+ \pi^-$

CP eigenstate.

Experimentally clean:

$$\pi^+ \pi^- \quad e^+ e^-$$

$$\pi^+ \pi^- \quad \mu^+ \mu^-$$



CP eigenvalue for $J/\psi \quad K_S^0$?



$$e^+ e^- \rightarrow B^0 \bar{B}^0$$

$$J = 1 \text{ for } \gamma$$

$$J_{\gamma}^{PC} = 1^{--}$$

B mesons $S_B = 0$.

$$e^+ e^- \rightarrow B^0 \bar{B}^0$$

$$J = 1$$

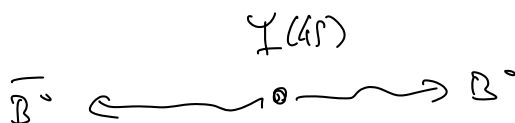
$$S = 0$$

$$0 \quad 0$$

$$\leftarrow e^+ \quad e^- \rightarrow$$

$$\Rightarrow L_{B^0 \bar{B}^0} = 1$$

orbital angular momentum in the final state.



Final state

$C = -1$ in the final state.

$$J = L_F + S_F$$

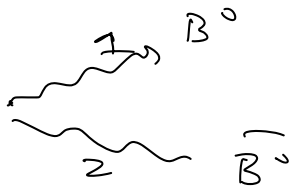
$$L_F = 1 \quad S_F = 0$$

$$e^+e^- \rightarrow b\bar{b} \quad C=-1 \text{ for } b\bar{b} \text{ pair from QFT}$$

$$b\bar{b} \text{ hadronize} \Rightarrow B^0 \bar{B}^0 \quad \tau_B = 1.57 \text{ fs}$$

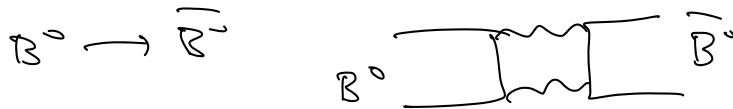
$$\text{Assume } \tau_B \rightarrow \infty$$

$$t=0 \quad C=-1$$



$$|\Psi_{\text{final state}}\rangle = a |B_1^0\rangle |\bar{B}_2^0\rangle + b |\bar{B}_1^0\rangle |B_2^0\rangle$$

neutral mesons can oscillate through weak interaction.



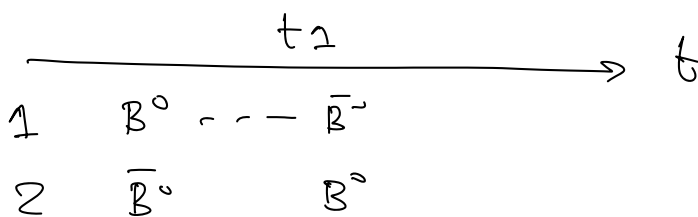
$$C=-1 \Rightarrow \text{there should always be } 1 B^0, 1 \bar{B}^0$$

$$|\Psi_{\text{final state}}\rangle = \frac{1}{\sqrt{2}} (|B^0\rangle |\bar{B}^0\rangle - |\bar{B}^0\rangle |B^0\rangle)$$

Entangled state

before the decay of 1 of the two mesons.

always 1 B^0 and 1 $\bar{B}^0$



Entanglement until
decay of 1 meson

$$e^+e^- \rightarrow B^0 \bar{B}^0$$

$\begin{array}{l} \searrow \text{J/\psi Ks} \\ \searrow \underline{D^- \pi^+} \\ \text{can only be } B^0 \end{array}$

$\begin{array}{l} \text{CP eigenstate accessible} \\ \text{to } B^0 \bar{B}^0 \end{array}$

Experimentally:

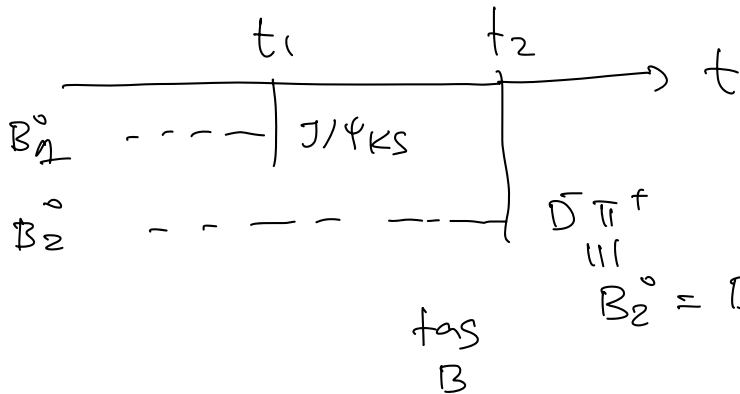
Final state

$$J/\psi K_S + D^- \pi^+$$

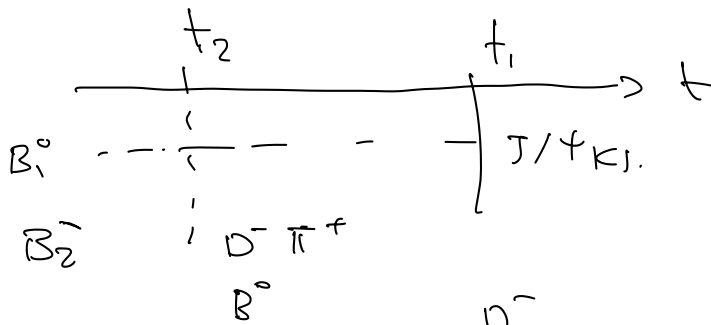
$$J/\psi K_S + D^+ \pi^-$$

B_{CP}

B_{tag}

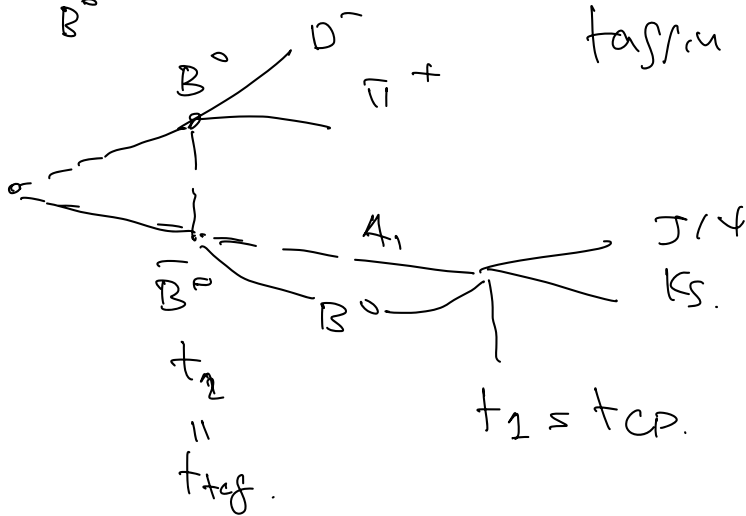


$$\Delta t = t_2 - t_1 < 0$$



$$\Delta t = t_2 - t_1 > 0$$

tagged B meson.



$$A_1: \bar{B}^0 \rightarrow J/\psi K_S$$

$$A_2: \bar{B}^0 \rightarrow B^0 \rightarrow J/\psi K_S$$

($J/\psi K_S$) with B^0 tag

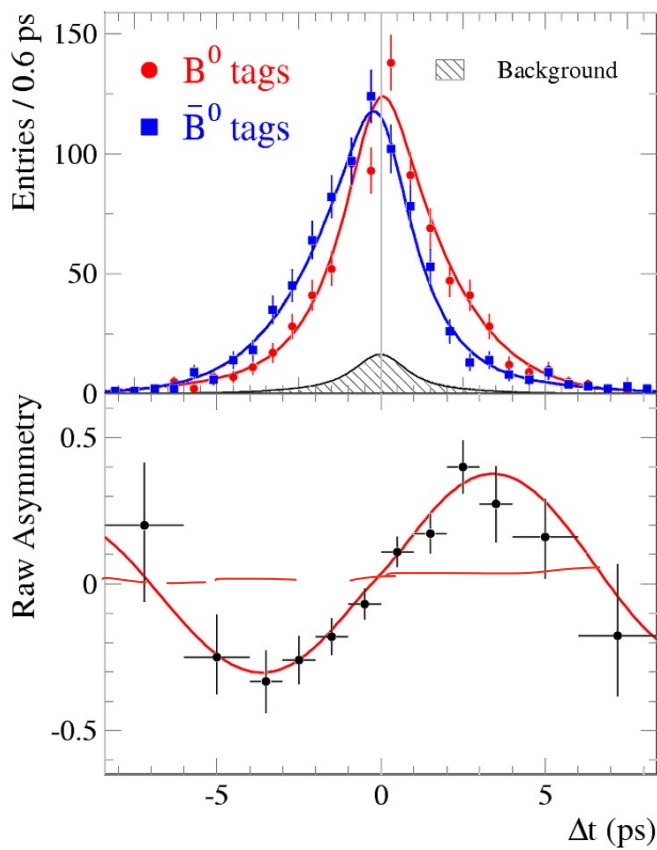
$$B_{tag}^0 \rightarrow D^- \pi^+$$

($J/\psi K_S$) with $\bar{B}^0$ tag

$$\bar{B}^0 \rightarrow D^+ \pi^-$$

$$A(B_1 \rightarrow J/\psi K_S; B_2 = \overset{B^0}{\underset{tagged}{D^- \pi^+}}) = A_1(\bar{B}^0 \rightarrow J/\psi K_S) + A_2(\bar{B}^0 \rightarrow B^0 \rightarrow J/\psi K_S)$$

$$\#(J/\psi K_S, B^0 tag) \propto |A_1|^2 + |A_2|^2 + \underbrace{A_1^\dagger A_2 + A_2^\dagger A_1}_{\sin 2\beta}$$



$$\Delta t = t_{\text{tag}} - t_{\text{CP}}$$

$$t_{\text{tag}}: \begin{aligned} B^0 &\rightarrow D^- \pi^+ \\ \bar{B}^0 &\rightarrow D^+ \pi^- \end{aligned}$$

$$t_{\text{CP}}: B \rightarrow J/\psi K_S$$

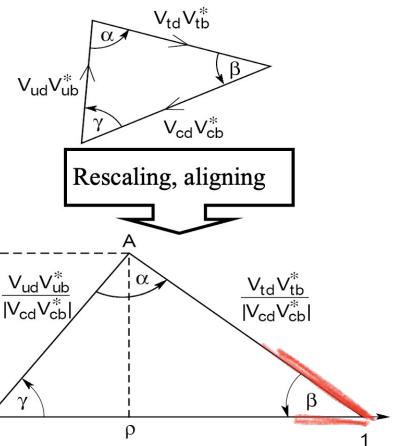
$$\# B^0 \text{ tags} \propto e^{-\Gamma t} (1 + \sin 2\beta \sin \Delta m \Delta t)$$

$$\# \bar{B}^0 \text{ tags} \propto e^{-\Gamma t} (1 - \sin 2\beta \sin \Delta m \Delta t)$$

$$A_{\text{Sym}} = \frac{\# B^0 \text{ tags} - \# \bar{B}^0 \text{ tags}}{\# B^0 \text{ tags} + \# \bar{B}^0 \text{ tags}}$$

2002

$$\sin 2\beta = 0.741 \pm 0.067_{(\text{stat})} \pm 0.033_{(\text{syst})}$$



$$\beta \neq 0$$

$$\Rightarrow \beta \neq 0$$

$$\Rightarrow \text{unitarity triangle non degenerate.}$$

$$\Rightarrow \text{CKM matrix has complex phase}$$

$$\Rightarrow \text{K-M. mechanism is correct}$$

$$\frac{dN}{dt} = \sigma \cdot \frac{dN_p}{dt} \text{ u.b.-d.}$$

$$\# (e^+e^- + \mu^+\mu^- + \tau^+\tau^- + \underbrace{\gamma\gamma + u\bar{u} + d\bar{d} + s\bar{s} + c\bar{c}}_{\text{hadrons}})$$

$$\sigma = \underbrace{\sigma_{e^+e^-} + \sigma_{\mu^+\mu^-} + \sigma_{\tau^+\tau^-} + \sigma_{\gamma\gamma}}_{\text{light}} + \underbrace{\sigma_{\gamma\bar{u}} + \sigma_{\gamma\bar{d}} + \sigma_{\gamma\bar{s}} + \sigma_{\gamma\bar{c}}}_{\text{hadrons}}$$

